Lab experiment

1.Create a new process by invoking the appropriate system call. Get the process identifier of the currently running process and its respective parent using system calls and display the same using a C program.

Aim:

To create a new process using the `fork()` system call and obtain the process ID (PID) and parent process ID (PPID) using `getpid()` and `getppid()` system calls.

Algorithm:

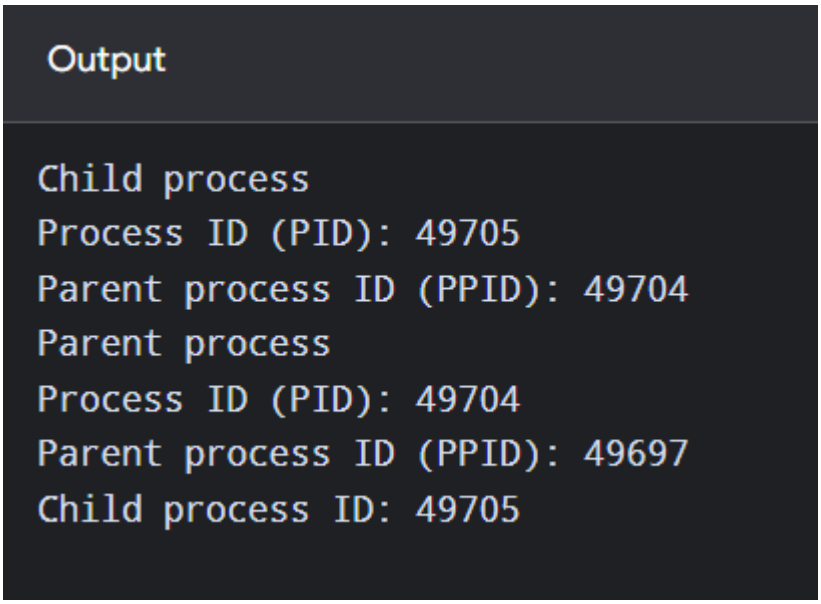
1. Include necessary headers: Include the necessary header files like <stdio.h> and <unistd.h> for using system calls.
2. Declare variables: Declare variables to hold the process ID (pid_t pid)and the parent process ID (pid_t ppid).
3. Get the current process ID: Use the getpid() system call to retrieve theprocess ID of the current process.
4. Get the parent process ID: Use the getppid() system call to retrieve theparent process ID.
5. Display the process IDs: Print the process IDs to the console.
6. Create a new process: Use the fork() system call to create a new process. Check the return value of fork() to determine whether the code is running in the parent or child process.
7. Display process IDs in child and parent processes: Depending on whether the process is the parent or the child, display the respective processIDs.

Coding:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
   pid_t pid;

   pid = fork();

   if (pid < 0)
   {
      printf("Process creation failed\n");
```

```
        return 1;
    }
    else if (pid == 0)
    {
        printf("Child process\n");
        printf("Process ID (PID): %d\n", getpid());
        printf("Parent process ID (PPID): %d\n", getppid());
    }
    else
    {
        wait(NULL);
        printf("Parent process\n");
        printf("Process ID (PID): %d\n", getpid());
        printf("Parent process ID (PPID): %d\n", getppid());
        printf("Child process ID: %d\n", pid);
    }

    return 0;
}
```

Output:



Result:

Thus, a new process was created using the `fork()` system call, and the process ID and parent process ID were obtained and displayed using `getpid()` and `getppid()`.